

## Section 3.5

68.

a.  $X$  has a hypergeometric distribution. It satisfies  $h(x; 6, 12, 20)$   
 $n=6$ .  $N=20$ .  $M=12$ .

$$b. P(X=2) = h(2; 6, 12, 20) = \frac{\binom{12}{2} \binom{8}{4}}{\binom{20}{6}} = 0.1192$$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) = \sum_{x=0}^2 h(x; 6, 12, 20)$$

$$= \frac{\binom{12}{0} \binom{8}{6}}{\binom{20}{6}} + \frac{\binom{12}{1} \binom{8}{5}}{\binom{20}{6}} + \frac{\binom{12}{2} \binom{8}{4}}{\binom{20}{6}}$$

$$= 0.0007 + 0.0174 + 0.1192$$

$$= 0.1373$$

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - P(X=0) - P(X=1)$$

$$= 1 - 0.0007 - 0.0174$$

$$= 0.9819$$

$$c. p = \frac{M}{N} = \frac{12}{20} = 0.6$$

$$E(X) = np = 3.6$$

$$V(X) = \frac{N-n}{N-1} np(1-p) = \frac{14}{19} \times 3.6 \times 0.4 = 1.061$$

$$\sigma = \sqrt{V(X)} = 1.030$$

69. a. The  $X$  satisfies the pmf  $h(x; 6, 7, 12) = \frac{\binom{7}{x} \binom{5}{6-x}}{\binom{12}{6}}$

$$P(X=5) = h(5; 6, 7, 12) = \frac{\binom{7}{5} \binom{5}{1}}{\binom{12}{6}} = 0.114$$

$$b. P(X \leq 4) = 1 - P(X \geq 5) = 1 - P(X=5) - P(X=6)$$

$$= 1 - \frac{\binom{7}{5} \binom{5}{1}}{\binom{12}{6}} - \frac{\binom{7}{6} \binom{5}{0}}{\binom{12}{6}}$$

$$= 1 - 0.114 - 0.007$$

$$= 0.879$$

$$c. p = \frac{M}{N} = \frac{7}{12} = 0.583$$

$$E(X) = np = 3.5$$

$$\sigma = \sqrt{V(X)} = \sqrt{\frac{N-n}{N-1} np(1-p)} = \sqrt{0.795} = 0.892$$

$$\text{Thus } P(X > \mu + \sigma) = P(X > 4.392) = P(X \geq 5)$$

$$= P(X=5) + P(X=6)$$

$$= 0.114 + 0.007$$

$$= 0.121$$

$$d. \text{ Since } n=15, N=400, \frac{n}{N} = 3.75\% < 5\%$$

We can see it as binomial distribution.  $p = \frac{M}{N} = 0.1$

It satisfies  $b(x; 15, 0.1)$

Thus  $P(X \leq 5) = B(5; 15, 0.1) = 0.998$ .

72. a. Let  $X$  = the number of the top four candidates are interviewed on the first day.  $X$  satisfies

$$\text{pmf } h(x; 6, 4, 11) = \frac{\binom{4}{x} \binom{7}{6-x}}{\binom{11}{6}}$$

b.  $p = \frac{M}{N} = \frac{4}{11} = 0.367$

$$E(X) = np = 2.18$$

75. a. Let  $X$  = the number of male children until there are two female children.  $X$  satisfies negative binomial distribution. The pmf is  $nb(x; 2, 0.5)$

$$nb(x; 2, 0.5) = \binom{x+1}{1} 0.5^2 \cdot (1-0.5)^x = (x+1) 0.5^{x+2}$$

b. That is,  $x = 4 - 2 = 2$ .

$$nb(2; 2, 0.5) = (2+1) 0.5^{2+2} = 0.188$$

c.  $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$$= nb(0; 2, 0.5) + nb(1; 2, 0.5) + nb(2; 2, 0.5)$$

$$= 1 \times 0.5^2 + 2 \times 0.5^3 + 0.188$$

$$= 0.688$$

$$d. E(X) = \frac{r(1-p)}{p} = \frac{2 \times 0.5}{0.5} = 2.$$

Thus the children expected to have is 4.

## Section 3.6

79. a. Let  $F(x; \mu) = F(x; 5)$  be the cdf

$$P(X \leq 8) = \sum_{x=0}^8 p(x; 5) = e^{-5} \sum_{x=0}^8 \frac{5^x}{x!} = F(8; 5) = 0.932$$

$$b. P(X=8) = p(8; 5) = F(8; 5) - F(7; 5) = 0.065$$

$$c. P(9 \leq X) = 1 - P(X \leq 8) = 1 - F(8; 5) = 0.068$$

$$d. P(5 \leq X \leq 8) = F(8; 5) - F(4; 5) = 0.492$$

$$e. P(5 < X < 8) = F(7; 5) - F(5; 5) = 0.251$$

84 a. Let  $X$  be the number of computers in the sample that have the defect. It satisfies the pmf  $b(x; 10000, 0.001)$

$$\text{Thus } E(X) = np = 10. \quad \sigma = \sqrt{V(X)} = \sqrt{np(1-p)} = 3.16$$

b. Since  $n > 100$ ,  $p < 0.01$ ,  $np < 20$ , the distribution can be approximately a Poisson distribution.

$$\mu \approx np = 10$$

$$P(X > 10) = 1 - F(10; 10) = 0.417$$



$$c. P(X=0) = \frac{e^{-10} \cdot 10^0}{0!} = e^{-10} = 4.54 \times 10^{-5}$$

86. a. Let  $X$  be the number of arrivals.

the distribution satisfies  $p(X; 5)$ .

$$p(4; 5) = \frac{e^{-5} \cdot 5^4}{4!} = 0.175$$

$$b. P(X \geq 4) = 1 - P(X \leq 3) = 1 - F(3; 5) = 0.735$$

$$c. E(X) = \mu = \lambda t = 5 \cdot \frac{45}{60} = 3.75$$

87. a.  $E(X) = \mu = \lambda t = 4 \times 2 = 8$ .

Thus the probability is  $p(10; 8) = \frac{e^{-8} 8^{10}}{10!} = 0.099$

$$b. \mu = \lambda t = 4 \times \frac{30}{60} = 2$$

The probability is  $p(0; 2) = \frac{e^{-2} \cdot 2^0}{0!} = 0.135$

$$c. E(X) = \mu = 2$$

